The Unified Theory of Local and Cosmic Expansion: Energy Volume Collapse and Black Hole Evaporation

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Abstract

This research presents a novel unified framework that replaces the concepts of Dark Matter and Dark Energy with the physical mechanisms of Black Hole dynamics. We propose that galaxy nucleus expansion (Local Expansion) and cosmic acceleration are results of "Energy Volume Collapse" in inflated stars and the subsequent evaporation of "Quark Ashes" and electrons from black holes.

1 Introduction

Modern cosmology relies on Dark Matter and Dark Energy to explain galaxy rotation and cosmic expansion. This paper proposes that these are not independent entities but manifestations of structural gravity and expended energy from black holes.

2 The Core Mechanism

2.1 Energy Volume vs. Material Volume

In this theory, 2/3 of an inflated star's volume is "Energy-based" (Gluon energy from nuclear shell distortion). Only 1/3 is physical matter. When a supernova occurs, the sudden collapse of this energy volume creates a localized vacuum, driving **Local Expansion**.

2.2 Structural Gravity

Gravity is defined as a structural physical mechanism rather than a field of energy. The rotation of black holes (50,000 km diameter) governs galaxy spiral identities, eliminating the need for Dark Matter.

3 Computational Framework

To validate the theory, a Python-based computational model was developed to simulate the creation of the local vacuum during the transition from an inflated star to a black hole.

```
# Lead Researcher: Wasfi Ahmad Mohammad Al-
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import math

class WasfiTheory:
    def __init__(self):
        self.ENERGY_RATIO = 0.66  # 2/3
            Gluon Energy
        self.BH_DIAMETER = 50000   # km

    def calculate_vacuum(self, radius_multi)
        :
        # Logic for Energy Volume Collapse
        # and Local Expansion creation
        pass
```

Listing 1: Proposed Computational Model

4 Conclusion

The "Funeral" of matter (evaporation of quarks and electrons) provides the expended energy for expansion. This framework suggests a self-contained universe where the lifecycle of stars regulates the fabric of space-time.